

SONAM GUPTA

AI & MACHINE LEARNING | WEB DEVELOPER | DATA ANALYST

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🎓 EDUCATION

Amrita Vishwa Vidyapeetham, Kerala Oct 2022 – Jun 2026
B.Tech in Computer Science (Artificial Intelligence) CGPA: 7.54
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, Computer Vision, Cloud Computing, Software Engineering.

💼 INTERNSHIP

NullClass Jan 2025 – Mar 2025
Machine Learning Intern

- Engineered three computer vision applications including face-recognition attendance, age prediction and vehicle color detection using TensorFlow, OpenCV and CNN architectures.
- Built complete machine learning pipelines covering data preprocessing, augmentation, model training and hyperparameter tuning, improving validation accuracy by nearly 15%.
- Optimized inference using transfer learning techniques, enabling low-latency real-time predictions suitable for production deployment.

📄 PUBLICATION

Visually Impaired Assistance Using Deep Learning Jul 2025
IEEE ICCCNT 2025, IIT Indore

- Developed an assistive system combining YOLO object detection, OCR and Text-to-Speech to improve navigation and environmental awareness for visually impaired users.
- Achieved reliable object recognition across varying lighting conditions while maintaining real-time performance.
- Published and presented the research paper at IEEE ICCCNT 2025 hosted by IIT Indore.

🔗 PROJECTS

AI-Powered Legal Chatbot — Indian Law Assistant 🔗 Mar 2025 – Jun 2025

- Built an AI legal assistant using Flask, RAG and vector databases to deliver accurate, citation-backed responses for Indian legal queries.
- Designed a semantic retrieval pipeline that reduced hallucinations by nearly 40% while significantly improving response relevance.
- Optimized document indexing and search to achieve fast, low-latency legal information retrieval.

LinkZone — Full Stack Social Platform 🔗 Aug 2025

- Developed a LinkedIn-inspired networking platform supporting authentication, posts, likes, comments and user connections.
- Implemented secure JWT authentication, bcrypt password encryption and scalable REST APIs with PostgreSQL.
- Built using Next.js App Router, TypeScript and Prisma, delivering responsive UI and production-ready architecture.

Omniplex — GenAI Chat & Search Platform 🔗 Oct 2024

- Built a Perplexity-inspired AI platform integrating Google Gemini API with live Google Search for intelligent responses.
- Integrated Firebase Authentication, Stripe payments and persistent chat history for a secure user experience.
- Designed a modular architecture enabling scalable AI chat, search and future plugin extensions.

⚙️ TECHNICAL SKILLS

🔗 **Languages**
Python, Java, C++, JavaScript, HTML, CSS, Scala, Typescript

📦 **Frameworks**
React, Next.js, Node.js, Express, Flask

🧠 **AI / ML**
Machine Learning, Deep Learning, NLP, Computer Vision

🗄️ **Databases**
PostgreSQL, MongoDB, MySQL

🔧 **Tools & Platforms**
Git, AWS, Azure, Power BI, Tableau, Jupyter Notebook

🌐 **Other Languages**
English, Hindi, Nepali, Bhojpuri

🏆 ACHIEVEMENTS

- Published Research**
Presented a research paper on AI-assisted navigation for the visually impaired at IEEE ICCCNT 2025, IIT Indore.
- Competitive Programming**
Qualified for Round 2 of the Unstop ICPC Challenge hosted by IIIT Nagpur (2024).
- Code360 – Coding Ninjas** ↗️
Earned multiple coding badges through DSA and programming challenges. Regularly practiced algorithmic problems to strengthen problem-solving and coding skills.
- Professional Certifications**
Earned 5+ certifications from Kaggle, DataCamp and ISKIMA, including Certified Professional Data Analyst (96%), covering Machine Learning, Deep Learning and Data Analytics.
- Virtual Internship**
Completed the Deloitte Forage Virtual Internship in Forensic Technology.
- Community Outreach**
Led the "Wellness Journey" SSR project, promoting healthy habits through interactive sessions and yoga.